Rusty Taylor

rdtaylorjr@gatech.edu rdtaylorjr.com github.com/rdtaylorjr ORCID 0009-0007-0702-2375
Colorado Springs, Colorado, USA

Research Interests

Computational textual criticism of the Hebrew Bible, eclectic digital critical editions, scribal practice, translation technique, and linguistic variation across ancient witnesses. Current work on the Psalms.

Education

M.S. Computer Science Georgia Institute of Technology 2025
Specialization in Artificial Intelligence GPA 4.0

B.S. Computer Science Western Governors University 2020
Excellence Award in Artificial Intelligence

M.A. Old Testament and Semitic Languages Trinity Evangelical Divinity School 2015
Magna cum laude GPA 3.83
Thesis: **"Translated Repetitions in Septuagint Proverbs"** readers Eric J. Tully and Dennis R. Magary. ProQuest 1597036

B.A. Biblical Studies Florida College 2005
Cum laude GPA 3.5 Minor in Mathematics

Languages

RESEARCH **Hebrew** Epigraphic, Biblical, Qumranic, Mishnaic **Greek** Hellenistic **Aramaic** Imperial, Jewish Literary, Syriac **Latin** **Coptic** Sahidic **Hittite**

MODERN **English** native **German** reading **French** reading **Hebrew** reading

Research Experience

Tehillim: Computational Analysis of the Psalms tehillim.dev github.com/rdtaylorjr/tehillim 2026–present
Collation of Judean Desert manuscripts and versions against MT. Statistical analysis of variants where multiple readings of consonants are possible. Hebrew poetry text-embedding benchmarks, similarity, and clustering.

Transcriber, Critical Edition of the Hebrew Psalter (CEHP) Duke Divinity School 2026–present
Transcribing MK31 (docID 350007) for the HBCE eclectic critical edition of Psalms 1–50, directed by Brent A. Strawn and Drew Longacre.

Graduate Research Assistant to Martin G. Abegg Jr. and Peter W. Flint Trinity Western University 2011–2013
Transcription and morphological tagging of Mishnah manuscripts. Syntactic tagging of Dead Sea Scrolls texts. Indexing of Naḥal Ḥever Greek papyri for The Dead Sea Scrolls Concordance, vol. 2 (Brill, 2016). Research and drafting for edited volumes with P. Flint.

3rd International Septuaginta Summer School Göttingen Academy of Sciences and Humanities 2012
"Current Issues in Septuagint Scholarship: New Insights and Debates. Readings in Proverbs."

Publications

CONFERENCE PAPERS

Taylor, Russell, Benjamin Herbert, and Michael Sana. **"Pun Intended: Multi-agent Translation of Wordplay with Contrastive Learning and Phonetic-Semantic Embeddings."** Pages 274–287 in *Experimental IR Meets Multilinguality, Multimodality, and Interaction.* Edited by Matthias Hagen et al. LNCS 17087. Springer, 2026. doi:10.1007/978-3-032-39150-6_20.

Taylor, Russell, Adam Brikman, and Prateek Awate. **"Searching for Sound-Meaning Collisions: Graph-Based Affordance Retrieval and Multi-Evaluator Ranking for Pun Translation at CLEF 2026 JOKER Task 2."** Pages 4295–4306 in *Working Notes of the Conference and Labs of the Evaluation Forum.* CEUR Workshop Proceedings 4283, 2026. **First place, JOKER Task 2.**

Taylor, Russell, Benjamin Herbert, and Michael Sana. **"Pun Intended: Multi-Agent Translation of Wordplay with Contrastive Learning and Phonetic-Semantic Embeddings for CLEF JOKER 2025 Task 2."** Pages 2870–2888 in *Working Notes of the Conference and Labs of the Evaluation Forum.* CEUR Workshop Proceedings 4038, 2025. **Best of CLEF 2025 Labs. First place, JOKER Task 2.**

"Translation Technique and Lexical Choice in Greek Exodus." Society of Biblical Literature Annual Meeting, IOSCS session, Chicago, November 2012.

"Syntactic Functions of the Particle כי in Qumran Hebrew." Pacific Northwest Regional Meeting of the AAR/SBL and ASOR, March 2012. First Prize, Graduate Student Paper Competition.

BOOK CHAPTERS

Taylor, Russell. **"'Your Eyes Will Be Opened': Inner-Biblical Allusion and Creation Language in John 9."** Pages 119–138 in *The Works No One Else Did: The Miracles of Jesus.* Edited by David McClister. Florida College Annual Lectures. Temple Terrace, FL: Florida College Press, 2019.

BOOK REVIEWS

Taylor, Russell. **Review of *Proverbs* (The Story of God Bible Commentary), by Ryan P. O'Dowd.** Books at a Glance, 2018.

Taylor, Russell. **Review of *Rediscovering the Dead Sea Scrolls: An Assessment of Old and New Approaches and Methods,* edited by Maxine L. Grossman.** Pages 455–458 in *Perspectives on Hebrew Scriptures X: Comprising the Contents of Journal of Hebrew Scriptures, vol. 13.* Edited by Christophe Nihan and Ehud Ben Zvi. Gorgias, 2017. doi:10.31826/9781463237646-034

Teaching

Instructional Associate, CS 6601 Artificial Intelligence Georgia Institute of Technology 2026–present
Wrote deep learning lecture series delivered by the professor of record. Assignment lead for a 1500-student graduate course, with responsibility for academic integrity review, student forums, office hours, and assessment.

Guest Lecturer, RELS 245 Elementary Biblical Hebrew Trinity Western University 2012–2013
Lectured on Biblical Hebrew grammar for a course taught by Martin G. Abegg Jr.

Professional Experience

Senior Machine Learning Engineer Spectrum 2024–present
Technical lead for real-time machine learning systems, responsible for model serving and the pipelines feeding them.

Senior Software Engineer Nielsen 2021–2024
Trained and evaluated classifiers over large-scale behavioral data; built the distributed pipelines supplying their training data.

Software Engineer Accordance Bible Software 2019–2021
Built and maintained specialized right-to-left Hebrew and Aramaic datasets optimized for linguistic analysis. Parsed and normalized unstructured ancient-language text for downstream processing.

Skills

DIGITAL PHILOLOGY	Text-Fabric, TEI P5 / IGNTP transcription, CollateX, eScriptorium/Kraken HTR
GENERATIVE AI	LangGraph, DSPy, MCP, OpenRouter, vLLM, TRL, Unsloth, Qdrant, lm-evaluation-harness
MACHINE LEARNING	PyTorch, Hugging Face (Transformers, PEFT/LoRA), sentence-transformers, scikit-learn
DATA ANALYSIS	NumPy, pandas, Polars, DuckDB, SciPy, statsmodels, matplotlib, Jupyter, Python

Memberships

Society of Biblical Literature, European Association of Biblical Studies, Association for Computational Linguistics